

Hemodialysis as adjunctive therapy for severe acetaminophen poisoning: a case report.

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Acetaminophen overdose is a common intoxication in daily practice the standard treatment is N-acetylcysteine (NAC) antidotal therapy for possible poisoning. However, dialysis procedures can remove the drug from the body effectively. We describe a case of acetaminophen overdose that was treated with both hemodialysis (HD) and NAC due to severe intoxication and slow drug clearance. A 37-year-old woman attempted suicide by ingestion of 100 tablets (500 mg each) of acetaminophen, and presented with vomiting, hematemesis and abdominal pain. The patient had elevated liver enzymes, coagulation defects, thrombocytopenia a high serum acetaminophen level (201 mg/l at 12 hours post-ingestion) with a prolonged half-life. Oral NAC was given; however, it was ineffective due to severe vomiting and hematemesis. HD as adjunctive therapy was initiated at 19 hours post-ingestion. HD reduced the serum acetaminophen level from 102.77 to 35.77 mg/l. Severe hepatic injury, bacteremia and pancytopenia were noted in the following days. The patient later recovered after treatment with NAC, HD and intensive supportive care. HD removed 66% of the total acetaminophen body burden during a single four-hour session, increased the clearance by 2.75-fold and shortened the half-life from 7.2 hours to 2.6 hours during HD. Through NAC therapy is the standard regimen for acetaminophen poisoning, in the severely poisoned patient who cannot tolerate NAC therapy, HD may be used as adjunctive therapy to enhance the elimination of acetaminophen.

Pattern of Paracetamol Poisoning: Influence on Outcome and Complications.

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Acute paracetamol poisoning due to a single overdose may be effectively treated by the early administration of N-acetylcysteine (NAC) as an antidote. The prognosis may be different in the case of intoxication due to multiple ingestions or when the antidote is started with delay. The aim of this work was to investigate the outcome of paracetamol poisoning according to the pattern of ingestion and determine the factors associated with the outcome. We performed a retrospective analysis over the period 2007–2017 of the patients who were referred to a tertiary hospital for paracetamol-related hepatotoxicity. Inclusion criteria were: accidental or voluntary ingestion of paracetamol, delay for NAC therapy of 12 h or more, liver enzymes (ALT) >1000 IU/L on admission. Ninety patients were considered. Poisoned patients following multiple ingestion were significantly older (45 ± 12 vs. 33 ± 14) ($p = 0.001$), with a higher incidence of liver steatosis ($p = 0.016$) or chronic ethanol abuse ($p = 0.04$). In comparison with the subgroup of favorable outcome, the patients with poor outcome were older, had higher values for ALT, bilirubin, lactate, and lower values for factor V and arterial pH. In multivariate analysis, the arterial lactate value was associated with a bad prognosis ($p < 0.02$) (adjusted odds ratio 1.74 and CI 95:1.09–2.77). The risk of poor outcome was greater in the subgroup with staggered overdose ($p = 0.02$), which had a higher mortality rate ($p = 0.01$). This retrospective analysis illustrates the different population patterns of patients who were admitted for a single ingestion of a paracetamol overdose versus multiple ingestions. This last subgroup was mainly represented by older patients with additional risk factors for hepatotoxicity; arterial lactate was a good predictor of severity.

Perioperative N-acetylcysteine: evidence and indications.

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Nonopioid analgesics serve to improve analgesia and limit side effects and risks of perioperative opioids. N-acetylcysteine (NAC), the primary treatment of acetaminophen toxicity, may have perioperative indications, including analgesia. NAC impacts glutathione synthesis, oxidant scavenging, glutamate receptor modulation and neuroinflammation. Potential perioperative benefits include arrhythmia prevention after cardiac surgery, decreased contrast-induced nephropathy, improved post-transplant liver function and superior pulmonary outcomes with general anesthesia. NAC may improve perioperative analgesia, with some studies displaying a reduction in postoperative opioid use. NAC is generally well tolerated with an established safety profile. NAC administration may predispose to gastrointestinal effects, while parenteral administration may carry a risk of anaphylactoid reactions, including bronchospasm. Larger randomized trials may clarify the impact of NAC on perioperative analgesic outcomes.

Concomitant use of activated charcoal and N-acetylcysteine.

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Activated charcoal is a safe, effective, inexpensive adjunct in the management of most toxic ingestions. It has the ability to adsorb a wide variety of drugs and chemicals, one of which is acetaminophen. N-acetylcysteine (NAC) is the specific antidote available for serious overdoses of acetaminophen. Current management of acetaminophen overdose, however, does not recommend the concomitant oral administration of these two useful agents because adsorption and inactivation of NAC by charcoal is believed to occur. Our study was designed to help evaluate the effect of activated charcoal on N-acetylcysteine absorption. Ten healthy male volunteers were each given in the first, or control, phase of the study an oral dose of 140 mg/kg NAC, and venous blood samples were obtained. In the second phase, after a washout period, each subject received 60 g activated charcoal orally followed immediately by 140 mg/kg NAC. NAC serum levels were measured using gas-liquid chromatography, and levels were compared with and without the concomitant administration of charcoal. Although only a small number of the subjects completed the study, the results showed that in both phases there were no significant differences in the peak NAC levels, the plasma half-life of NAC, or the calculated area under the curve. We recommend that NAC and activated charcoal not be used clinically until further studies are completed.

Hepatotoxicity despite early administration of intravenous N-acetylcysteine for acute acetaminophen overdose.

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The objective was to evaluate the effectiveness of intravenous N-acetylcysteine (IV NAC; 300 mg/kg over 21 hours) in early acute acetaminophen (APAP) overdose patients. This observational case series included patients hospitalized between 2004 and 2007 for acute APAP overdoses and who were reported to a regional poison center. Inclusion criteria were plasma APAP concentrations on or above the treatment line on the Rumack-Matthew nomogram, administration of IV NAC within 8 hours of ingestion, and follow-up to known outcome. The hospital chart of each patient who received IV NAC for longer than the standard 21 hours was reviewed. Hepatotoxicity was defined as hepatic aminotransferase levels

greater than 1,000 IU/L. Seventy-seven patients met inclusion criteria and received at least 21 hours of IV NAC for an acute APAP overdose. Seven patients received antidotal therapy for greater than 21 hours. These patients tended to have ingested combination preparations, have very high initial plasma APAP concentrations, and had persistently elevated plasma concentrations during their hospital stay. Hepatotoxicity occurred in 4 patients (5.2%, 95% confidence interval [CI] = 0.2% to 10.1%), including 1 death and 1 liver transplantation. Hepatotoxicity developed in 5.2% of cases, suggesting that the 21-hour IV NAC regimen is suboptimal in some patients. In addition to high initial plasma APAP concentrations, APAP product formulation and persistently elevated plasma APAP concentrations were identified as factors possibly associated with developing hepatotoxicity. The authors propose a tailored approach to the discontinuation of IV NAC and point out the need for reevaluation of optimal doses and duration of therapy.

Massive acetaminophen ingestion managed successfully with N-acetylcysteine, fomepizole, and renal replacement therapy.

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Acetaminophen ingestion is routinely managed with the antidote, N-acetylcysteine (NAC). Massive acetaminophen poisoning has been treated successfully with adjunctive therapies such as fomepizole and hemodialysis. Fomepizole functions by inhibiting cytochrome p560, which prevents tylenol from forming its toxic metabolite, NAPQI. Prior cases have demonstrated favorable outcomes and a significant drop in acetaminophen levels after a single session of intermittent hemodialysis and continuous veno-venous hemofiltration (CVVH). However, the recommended dosage adjustments of NAC and fomepizole while a patient is undergoing CVVH has not been well reported. We present a case of an 18-year-old male who presented after ingesting 125 g of tylenol. His 4-hour acetaminophen level was 738.6 µg/mL. He was treated with NAC, fomepizole, and a single 4-hour session of hemodialysis. His acetaminophen level remained elevated at 730 µg/mL despite the hemodialysis session. CVVH was initiated, and he was given intravenous NAC at 12.5 mg/kg/h, oral NAC at 70 mg/kg every 4 hours, and intravenous fomepizole at 10 mg/kg every 6 hours. His tylenol levels became undetectable 57 hours after ingestion, and he did not develop permanent liver toxicity. This case encourages the use of CVVH for massive tylenol ingestion when a single run of intermittent hemodialysis is not effective in lowering the tylenol level. NAC, fomepizole, and CVVH can prevent unfavorable outcomes in massive acetaminophen ingestion when provided at an appropriate dose and frequency.

[Diagnostics and treatment of selected clinically relevant, acute drug intoxications].

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Acute drug poisoning due to accidental or self-damaging overdoses is responsible for 5-10% of emergency medical interventions in Germany. The treatment of asymptomatic to life-threatening courses requires extensive expertise. On the basis of a selective literature search, this article gives an overview of selected clinically relevant, acute drug poisonings with regard to epidemiology, symptomatology, diagnostics, and therapy. Intoxications with psychotropic drugs are the most common drug intoxications. Poisoning with tricyclic antidepressants causes anticholinergic, central nervous, and cardiovascular symptoms. Less toxic are selective serotonin reuptake inhibitors (SSRIs); the intoxication may be

characterized by serotonin syndrome. Malignant neuroleptic syndrome is a severe complication of neuroleptic poisoning. Poisoning with analgesics is clinically relevant due to its high availability. For paracetamol poisoning, intravenous acetylcysteine is available as an antidote. Hemodialysis may be indicated for severe salicylate intoxication. Poisoning with nonsteroidal anti-inflammatory drugs is usually only associated with mild signs of intoxication. Poisoning with cardiac drugs (β -blockers and calcium antagonists) can cause life-threatening cardiovascular events. In addition to symptomatic therapy, insulin glucose therapy also plays an important role. The majority of acute drug poisonings can be treated adequately by symptomatic and partly intensive care therapy - if necessary with the application of primary and secondary toxin elimination. Depending on the severity of the intoxication, pharmacology-specific therapy must be initiated.

Effects of intra-articular botulinum toxin type A (Botox) in dogs with chronic osteoarthritis.

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To evaluate the effects of intra-articular botulinum neurotoxin type A (BoNT/A) in dogs with chronic osteoarthritis. Client-owned dogs with lameness and discomfort attributed to unilateral elbow or hip osteoarthritis were eligible for inclusion ($n = 5$). All dogs had BoNT/A (25 units) administered to the affected joint (2 elbows, 3 hips). Dogs were evaluated by pressure platform gait analysis before and at two, four, eight, and 12 weeks post-injection, and by client perception of outcome. In experimental limbs, ground reaction forces (peak vertical force and vertical impulse) consistently improved for a variable period of time following intra-articular BoNT/A therapy. These changes were not, however, observed in the contralateral limbs, in which values remained relatively unchanged or decreased. Four out of five owners reported at least some improvement in their dog's condition following treatment. A multimodal approach with the intra-articular administration of BoNT/A may be an option for osteoarthritis patients that are unresponsive to medical management and unable to undergo surgery. However, the findings of this study are preliminary and must be verified by further investigation.

Mesotherapy: cosmetic applications.

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Mesotherapy, which has been compared to the anti-wrinkle injection Botox, has only recently become popular in the United States for providing body contouring and spot weight loss. Most adverse reactions associated with mesotherapy are mild and transient, and supporters of the procedure consider it a safer alternative to liposuction. Mainly owing to the absence of safety and efficacy data pertaining to mesotherapy, liposuction is currently the only method for fat removal that is endorsed by the American Society of Plastic Surgeons. Although mesotherapy has been used as a treatment for pain syndromes, arthritis, and many other disorders, this article focuses on its cosmetic applications.

Intra-Articular Injection of Botulinum Toxin for the Treatment of Knee Osteoarthritis: A Systematic Review of Randomized Controlled Trials.

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The purpose of the present paper was to review the available evidence on intra-articular botulinum toxin (BTX) injection in the treatment of knee osteoarthritis and to compare it to other conservative treatment options. A systematic review of the literature was performed on the PubMed, Scopus, Cochrane Library, Web of Science, Pedro and Research Gate databases with the following inclusion criteria: (1) randomized controlled trials (RCTs), (2) written in the English language, and (3) published on indexed journals in the last 20 years (2001-2021) dealing with the use of BTX intra-articular injection for the treatment of knee OA. The risk of bias was assessed using the Cochrane Risk of Bias tool for RCTs. Nine studies involving 811 patients in total were included. Patients in the control groups received different treatments: conventional physiotherapy, hyaluronic acid injection or prolotherapy or a combination thereof in 5 studies, steroid infiltrative therapy (triamcinolone) in 1 study, placebo in 2, and local anesthetic treatment in 1 study. Looking at the quality of the available literature, two of the included studies reached "Good quality" standard, three were ranked as "Fair", and the rest were considered "Poor". No major complications or serious adverse events were reported following intra-articular BTX, which provided encouraging pain relief, improved motor function, and quality of life. Based on the available data, no clear indication emerged from the comparison of BTX with other established treatments for knee OA. The analysis of the available RCTs on BTX intra-articular injection for the treatment of knee OA revealed modest methodological quality. However, based on the data retrieved, botulinum toxin has been proven to provide good short-term outcomes, especially in patients with pain sensitization, by modulating neurotransmitter release, peripheral nociceptive transduction, and acting on the control of chronic pain from central sensitization.